

Write your questions
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Comparison Test

Let $0 < a_n \leq b_n$ for all n .

If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$

If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$

Determine if the following series converge or diverge.

1. $\sum_{n=1}^{\infty} \frac{1}{3 + 2^n}$

2. $\sum_{n=2}^{\infty} \frac{1}{4^n - 3}$

3. $\sum_{n=1}^{\infty} \frac{1}{7n^2 + 4}$

Limit Comparison Test

If $a_n > 0$, $b_n > 0$ and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$ (where L is finite and positive), then

$$\sum_{n=1}^{\infty} a_n \text{ and } \sum_{n=1}^{\infty} b_n$$

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Determine if the following series converge or diverge.

4.
$$\sum_{n=1}^{\infty} \frac{2n^2 - 2}{5n^5 + 3n + 1}$$

5.
$$\sum_{n=1}^{\infty} \frac{1}{5n^2 + 5n + 5}$$

6.
$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{3n+2}}$$

7.
$$\sum_{n=1}^{\infty} \frac{n^3 - 7}{2n^5 + n^2 + n + 1}$$

8.
$$\sum_{n=1}^{\infty} \frac{n3^n}{4n^3 + 2}$$

9.
$$\sum_{n=1}^{\infty} \frac{n}{\sqrt{n^3 + n}}$$

10.6 Comparison Tests for Convergence

Calculus

Practice

1. Which of the following statements about convergence of the series $\sum_{n=1}^{\infty} \frac{1}{\ln(n+2)}$ is true?

- (A) $\sum_{n=1}^{\infty} \frac{1}{\ln(n+2)}$ converges by comparison with $\sum_{n=1}^{\infty} \frac{1}{n}$
- (B) $\sum_{n=1}^{\infty} \frac{1}{\ln(n+2)}$ converges by comparison with $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3}}$
- (C) $\sum_{n=1}^{\infty} \frac{1}{\ln(n+2)}$ diverges by comparison with $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3}}$
- (D) $\sum_{n=1}^{\infty} \frac{1}{\ln(n+2)}$ diverges by comparison with $\sum_{n=1}^{\infty} \frac{1}{n}$
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2. Which of the following series converges?

(A) $\sum_{n=1}^{\infty} \frac{3n}{n^3 + 2}$

(B) $\sum_{n=1}^{\infty} \frac{5n}{2n + 1}$

(C) $\sum_{n=1}^{\infty} \frac{7n}{n^2 + 1}$

(D) $\sum_{n=1}^{\infty} \frac{5^n}{4^n + 1}$

3. Use the Comparison Test to determine whether the series $\sum_{n=1}^{\infty} \frac{1}{2 + 5^n}$ converges or diverges.

4. Which of the following series can be used with the Limit Comparison Test to determine convergence of the series $\sum_{n=1}^{\infty} \frac{n^3}{n^4 + 3}$?

(A) $\sum_{n=1}^{\infty} \frac{n}{n+3}$

(B) $\sum_{n=1}^{\infty} \frac{1}{n^3 + 3}$

(C) $\sum_{n=1}^{\infty} \frac{1}{n}$

(D) $\sum_{n=1}^{\infty} \frac{1}{n^4}$

5. Consider the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$, where $a_n > 0$ and $b_n > 0$ for $n \geq 1$. If $\sum_{n=1}^{\infty} a_n$ diverges which of the following must be true?

(A) If $a_n \leq b_n$, then $\sum_{n=1}^{\infty} b_n$ converges.

(B) If $a_n \leq b_n$, then $\sum_{n=1}^{\infty} b_n$ diverges.

(C) If $b_n \leq a_n$, then $\sum_{n=1}^{\infty} b_n$ converges.

(D) If $b_n \leq a_n$, then $\sum_{n=1}^{\infty} b_n$ diverges.

6. Consider the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$, where $a_n > 0$ and $b_n > 0$ for $n \geq 1$. If $\sum_{n=1}^{\infty} b_n$ converges which of the following must be true?

(A) If $a_n \leq b_n$, then $\sum_{n=1}^{\infty} a_n$ diverges.

(B) If $a_n \leq b_n$, then $\sum_{n=1}^{\infty} a_n$ converges.

(C) If $b_n \leq a_n$, then $\sum_{n=1}^{\infty} a_n$ diverges.

(D) If $b_n \leq a_n$, then $\sum_{n=1}^{\infty} a_n$ converges.

7. Let $a > 0$, $b > 0$, and $c > 0$. Determine whether the series $\sum_{n=0}^{\infty} \frac{1}{an^2 + bn + c}$ converges or diverges.

8. Determine the convergence or divergence of the series $\sum_{n=2}^{\infty} \frac{1}{6^n + 6}$.

9. For the series $\sum_{n=1}^{\infty} \frac{n3^n}{2n^4 - 2}$, which of the following could be used with the Limit Comparison Test?

(A) $\sum_{n=1}^{\infty} \frac{1}{n^4}$

(B) $\sum_{n=1}^{\infty} \frac{3^n}{n^4}$

(C) $\sum_{n=1}^{\infty} \frac{1}{n^3}$

(D) $\sum_{n=1}^{\infty} \frac{3^n}{n^3}$

10. Which of the following can be used with the Comparison Test to determine the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{2 + \sqrt{n}}$?

(A) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

(B) $\sum_{n=1}^{\infty} \frac{1}{n}$

(C) $\sum_{n=1}^{\infty} \frac{1}{n^2}$

(D) $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$

11. Which of the following series diverge?

I. $\sum_{n=1}^{\infty} \frac{\sqrt[3]{n}}{n}$

II. $\sum_{n=1}^{\infty} \frac{1}{n^3 - 27}$

III. $\sum_{n=1}^{\infty} \frac{1}{4^n + 1}$

(A) I only

(B) II only

(C) I and II only

(D) I, II, and III

12. Consider the series $\sum_{n=2}^{\infty} \frac{1}{n^p \ln n}$, where $p \geq 0$. For what values of p is the series convergent?

13. Determine whether the series $\sum_{n=1}^{\infty} \frac{n-3}{n^3}$ converges or diverges.

14. Consider the series $1 + \frac{1}{5} + \frac{1}{9} + \frac{1}{13} + \cdots = \sum_{n=1}^{\infty} \frac{1}{4n-3}$. Use the Limit Comparison Test with the series $\sum_{n=1}^{\infty} \frac{1}{4n}$ to determine the convergence of the series.

10.6 Comparison Tests for Convergence

15. Consider the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$, where $a_n > 0$ and $b_n > 0$ for $n \geq 1$. If $a_n \leq b_n$, then which of the following must be true?

- (A) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ diverges.
- (B) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges.
- (C) If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.
- (D) If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ converges.

16. Consider the series $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$, where $a_n > 0$ and $b_n > 0$ for $n \geq 1$. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 2$, then which of the following must be true?

- I. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ diverges.
- II. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ converges.
- III. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges.
- IV. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.

- (A) I only
- (B) II only
- (C) III only
- (D) IV only
- (E) I and II only
- (F) III and IV only